CancerOS — Medical Imaging & Digital Twin Specification

Document ID: COS-SPEC-005
Version: 1.0
Status: Proposed engineering specification
Date: October 22, 2026
Parent documents: COS-PRD-001, COS-SPEC-003, COS-SPEC-004
Subsystem: Medical Imaging, Tumor Reconstruction and Digital Twin Engine
Clinical scope: Breast cancer
Primary imaging modality: Dynamic contrast-enhanced breast MRI (DCE-MRI)

---

1. Purpose

This specification defines how CancerOS converts real breast MRI studies into geometrically accurate, measurable, interactive tumor representations and how those representations are linked across genuine and simulated longitudinal observations.

The imaging subsystem shall support:

1. Authentic source image ingestion.
2. MRI integrity and spatial metadata validation.
3. Medical imaging preprocessing.
4. Tumor localization and segmentation.
5. Three-dimensional tumor geometry reconstruction.
6. Physical tumor measurements.
7. Longitudinal imaging alignment.
8. Tumor identity and measurement tracking.
9. Synthetic follow-up tumor geometry.
10. Interactive web-based 3D visualization.
11. Provenance and uncertainty disclosure.
12. Quantitative imaging evaluation.

The system shall prioritize scientifically defensible results over visually dramatic but unsupported representations.

2. Product Boundary

2.1 Included

- Breast-region MRI.
- Source DICOM images.
- Processed NIfTI imaging volumes.
- Imported tumor segmentation masks.
- Automated segmentation proposals where supported.
- Three-dimensional tumor mesh generation.
- Tumor diameter and volume calculation.
- Image registration across appropriate examinations.
- Time-indexed lesion measurements.
- Synthetic tumor geometry derived from authentic baselines.
- Interactive 3D viewing and optional synchronized 2D slices.
- Imaging provenance and validation.

2.2 Excluded

- Whole-body anatomical reconstruction.
- Full organ physiological simulation.
- Cellular tumor microenvironment simulation.
- Drug diffusion modeling.
- Microscopic invasion modeling.
- Metastatic spread visualization.
- Histopathology image segmentation.
- Clinical diagnostic interpretation of imaging.
- Autonomous radiology reporting.
- Claims that a synthetic mesh is an authentic follow-up MRI.

- Radiotherapy treatment-planning functionality.

2.3 Core distinction

The digital twin contains measured, derived, or simulated representations of lesion geometry.

It shall not claim to reproduce all internal biological properties of the original cancer.

---

3. Imaging Data Sources

Primary candidates:

- I-SPY2 longitudinal breast MRI.
- I-SPY1 longitudinal breast MRI.
- Duke Breast Cancer MRI.
- BreastDCEDL processed MRI resources.

Each source shall be registered according to COS-SPEC-003.

3.1 Imaging-source eligibility

A study may be used only when its format and spatial geometry are sufficiently interpretable.

Minimum eligibility:

- Valid image dimensions.
- Known spatial spacing.
- Known image orientation or a documented conversion method.
- Usable source image pixels.
- Stable study and series identifiers.
- Relevant acquisition information.
- Permitted dataset use.
- A lesion annotation or applicable segmentation method for the intended task.

3.2 Annotation quality classes

CancerOS shall classify annotations as:

A — Expert or reference lesion segmentation: Delineated lesion mask whose intended anatomical meaning is established.

B — Validated model-derived segmentation: Automated mask with documented evaluation against a relevant reference.

C — Approximate lesion annotation: Bounding box, region of interest, or approximate mask with documented limitations.

D — Algorithmic analysis region: Region representing a separate construct, such as a functional tumor-volume enhancement criterion, which may not equal the full anatomical lesion.

E — No suitable lesion segmentation: Imaging may still be viewed, but lesion volume reconstruction is unavailable.

This distinction is mandatory.

A functional tumor-volume analysis mask must not automatically be relabeled as an expert anatomical tumor segmentation.

---

4. Medical Imaging Processing Architecture

The pipeline shall follow this sequence.

Original imaging

DICOM / approved source imaging

↓

Integrity validation

Study, series, metadata, geometry and pixel checks

↓

Volume construction

Sort and assemble slices in physical space

↓

Standardization

Orientation, spacing, intensity and sequence management

↓

Tumor localization

Associate lesion annotations or eligible segmentation model

↓

Segmentation and quality control

Generate or load mask; validate spatial correspondence

↓

Measurements

Physical volume, dimensions and mask statistics

↓

Geometry reconstruction

Three-dimensional lesion mesh and optional surrounding anatomy

↓

Temporal association

Link geometry and measurements to specific observation dates

↓

Digital twin publication

Store versioned artifacts for web visualization and AI interpretation.

Each stage shall create structured processing metadata.

Failed stages shall not result in fabricated or silently corrected clinical measurements.

---

5. MRI Ingestion Requirements

IMG-001 — Source format

The subsystem shall support DICOM imaging and approved processed NIfTI files.

IMG-002 — Study grouping

Images shall be grouped by verified source study and series identifiers.

IMG-003 — Acquisition separation

Different DCE contrast phases shall be represented as acquisitions within a study, not separate clinical visits.

IMG-004 — Slice ordering

Slices shall be ordered using image geometry rather than relying only on filenames or arbitrary slice-instance numbering.

IMG-005 — Physical coordinates

Pixel data shall retain physical spatial information.

IMG-006 — Source preservation

Original imaging files shall remain unchanged.

IMG-007 — Integrity checks

The system shall reject or quarantine:

- Unreadable images.
- Inconsistent dimensions without a documented conversion.
- Missing essential geometry.
- Invalid orientation.
- Nonfinite pixel data.
- Duplicate conflicting slices.
- Unsupported spatial arrangements.
- Series whose slices cannot be assembled reliably.

IMG-008 — Privacy

DICOM files must be inspected for patient-identifying metadata before any export or external processing.

No source imaging shall be transmitted to external AI services without verified authorization.

---

6. MRI Preprocessing Specification

6.1 Preprocessing objectives

Preprocessing shall make images suitable for reliable segmentation, measurement and visualization while preserving the relationship to the original physical imaging space.

6.2 Required stages

P-01 — Load source volume

Use SimpleITK, pydicom, or another validated medical imaging reader.

Record:

- Source artifact identifier.
- File checksum.
- Dimensions.
- Pixel type.
- Spacing.
- Origin.
- Orientation.
- Image affine or direction matrix.

P-02 — Validate spatial geometry

Confirm that the physical-space interpretation is consistent.

Do not infer physical dimensions from array shape alone.

P-03 — Standardize orientation

Convert to a documented internal orientation convention.

DICOM commonly uses LPS coordinates; NIfTI workflows commonly use RAS conventions.

Store and verify any coordinate transformation.

P-04 — Resample where required

For segmentation and model inference, resample using a model-compatible target spacing.

The target spacing shall be chosen based on data characteristics or the segmentation model's documented requirements.

Do not assume that 1 mm isotropic spacing is optimal for every breast MRI dataset.

Image intensity resampling may use continuous interpolation.

Categorical lesion masks must use nearest-neighbor interpolation or an appropriate label-preserving method.

P-05 — Normalize intensity

Use an explicitly documented intensity normalization method.

Potential techniques include:

- Robust percentile clipping.
- Foreground-aware z-score normalization.
- Model-specific normalization.
- DCE phase-specific preprocessing.

All actual normalization parameters shall be recorded.

MRI intensities are not inherently standardized quantitative tissue measurements.

P-06 — Select image channels

Identify the MRI sequence and DCE phase inputs expected by the segmentation method.

Precontrast and postcontrast images must not be mixed arbitrarily.

P-07 — Crop an appropriate region

Where anatomically justified, crop to a relevant breast or lesion region for efficient computation.

Retain transformation metadata so coordinates can be mapped back to the source image.

P-08 — Quality assessment

Detect likely processing failures, including:

- Missing image planes.
- Severe geometry inconsistencies.
- Invalid intensity ranges.
- Incorrect channel mapping.
- Mask-image misalignment.
- Empty or corrupted outputs.

6.3 Optional processing

Bias-field correction, motion correction, denoising or enhancement may be evaluated, but must not be applied blindly.

Each optional processing step requires a recorded purpose and evaluation of its impact on downstream measurements.

6.4 Output artifacts

- Standardized imaging volume.
- Spatial transformation metadata.
- Preprocessing configuration.
- Quality-control summary.
- Processed-image checksum.
- Original source reference.

---

7. Tumor Segmentation

7.1 Required capability

CancerOS must represent a tumor lesion using an appropriate voxel-level mask or validated geometric approximation.

7.2 Segmentation modes

Mode A — Reference-mask import (P0)

The system shall import and validate an existing suitable lesion mask.

This is the baseline functionality necessary to guarantee that the 3D demonstration can work without training a new model.

Mode B — Automated segmentation (P1, promoted to P0 only if validated and feasible)

Use a medical-imaging segmentation model to propose tumor boundaries.

Candidate methods:

- nnU-Net v2.
- MONAI-based U-Net.
- A documented breast MRI-specific pretrained segmentation checkpoint.

The existence of a framework does not mean a compatible, validated pretrained breast MRI model is available.

Mode C — Research correction workflow (P1)

Allow a permitted research user to inspect or correct segmentation results using supported editing tools.

Corrections must be versioned and attributed.

7.3 Model input

Inputs shall include:

- Valid processed MRI volume.
- Required imaging channels.
- Model-compatible spatial orientation.
- Expected voxel spacing.
- Preprocessing metadata.

7.4 Model output

Outputs shall include:

- Segmentation mask.
- Lesion class.
- Model identifier and version.
- Source study and series references.
- Inference configuration.
- Processing time.
- Quality and uncertainty indicators where supported.

7.5 Segmentation postprocessing

Permitted operations may include:

- Connected-component analysis.
- Removal of explicitly defined small artifacts.
- Mask geometry checks.

- Conversion to source space.
- Lesion component labeling.

Postprocessing must not silently remove actual lesions merely because they are smaller than a heuristic threshold.

7.6 Evaluation metrics

Report appropriate metrics on held-out reference annotations:

- Dice similarity coefficient.
- Intersection over Union.
- 95th-percentile Hausdorff distance in millimeters.
- Volume absolute error.
- Volume percentage error.
- Empty-mask failure rate.
- Processing success rate.

A Dice score alone is insufficient to establish clinically acceptable segmentation.

7.7 Segmentation status

Possible states:

- REFERENCE_IMPORTED
- AUTOMATED_PENDING
- AUTOMATED_COMPLETED
- QUALITY_REVIEW_REQUIRED
- APPROVED_FOR_RESEARCH
- REJECTED

These statuses describe processing and research usability, not approval for clinical diagnosis.

---

8. Tumor Measurement Engine

8.1 Primary measurement

The initial 3D anatomical measurement shall be lesion volume.

For a binary mask on a uniform physical voxel grid:

Volume_mm3 = number_of_tumor_voxels × absolute_voxel_volume_mm3

Where voxel volume is determined from the spatial affine or grid transformation.

For orthogonal grids, this commonly reduces to the product of the three voxel spacings.

Display conversion:

Volume_cm3 = Volume_mm3 / 1000

8.2 Additional measurements

Where supported:

- Maximum physical lesion diameter.
- Orthogonal extents.
- Lesion centroid.
- Bounding-box dimensions.
- Percentage volume change.
- Absolute volume change.
- Distance between lesion centroids after valid alignment.

Axis-aligned bounding-box length must not automatically be represented as the longest RECIST lesion diameter.

8.3 Longitudinal comparisons

For comparable measurements:

Absolute_change = Current_value - Baseline_value

Percent_change = 100 × (Current_value - Baseline_value) / Baseline_value

If the baseline denominator is zero or unavailable, percentage change shall not be calculated.

8.4 Measurement uncertainty

The system shall distinguish:

- Segmentation uncertainty.

- Registration uncertainty.
- Acquisition-related variability.
- Simulation uncertainty.

A derived value must not be presented with unjustified numerical precision.

8.5 Measurement reproducibility

The measurement engine must return consistent values for the same verified mask and physical geometry.

---

9. Three-Dimensional Tumor Reconstruction

9.1 Objective

Convert a valid voxel-level lesion mask into a rotatable 3D surface model suitable for browser visualization.

9.2 Processing method

The initial implementation may use an isosurface-extraction algorithm such as marching cubes.

Processing steps:

1. Load the validated segmentation mask.
2. Verify mask spatial metadata.
3. Extract the lesion surface.
4. Transform vertices into physical coordinates.
5. Recalculate normals.
6. Validate mesh integrity.
7. Apply controlled smoothing when needed.
8. Optionally simplify the mesh for browser performance.
9. Export a web-compatible geometry artifact.
10. Link it to the source segmentation and measurement record.

9.3 Mesh quality requirements

Check:

- Finite vertex coordinates.
- Valid face indices.
- No unexpected empty mesh.
- Consistent orientation.
- Acceptable geometry simplification.
- Valid physical dimensions.
- Traceability to the source mask.

9.4 Volume authority

The authoritative tumor volume shall generally be calculated from the validated mask, not inferred from a heavily smoothed or simplified display mesh.

Display mesh transformations must not silently change the stored clinical research measurement.

9.5 Anatomical surroundings

The viewer may include:

- Breast-region imaging slices.
- A breast-region surface where it is genuinely derived from source imaging.
- An illustrative translucent shell for orientation.

Any purely illustrative anatomy must carry that classification.

---

10. Longitudinal Imaging Registration

10.1 Objective

Where appropriate, align repeated real imaging studies to support meaningful spatial comparison.

10.2 Registration methods

Initial candidate methods:

- Rigid registration.
- Affine registration.

Deformable registration may be investigated later but is not mandatory for the MVP.

Breast deformation, treatment-associated changes, patient positioning and scan protocols can create substantial registration challenges.

10.3 Registration inputs

- Fixed reference image.
- Moving follow-up image.
- Source spatial metadata.
- Selected registration region.
- Chosen transformation class.
- Registration parameters.

10.4 Registration outputs

- Transformation matrix or transform object.
- Registered image artifact.
- Registration quality assessment.
- Source and destination study identifiers.
- Method/version information.

10.5 Quality checks

- Physical overlap.
- Anatomical alignment.
- Deformation or transformation plausibility.
- Landmark agreement where available.
- Visual review for curated demonstration cases.

10.6 Failure behavior

If registration quality is inadequate:

- The system shall preserve the source images.
- Independent volume measurements may still be displayed.

- The interface shall warn that spatial overlays are not reliable.
- Unsupported spatial movement claims shall be suppressed.

10.7 Tumor tracking and identity

Longitudinal tumor tracking requires two distinct operations:

1. Associate lesions across visits where identity is supportable.
2. Compare their measurements and positions where spatial alignment is reliable.

An overlap-based algorithm shall not automatically equate all neighboring lesions with the same biological lesion.

For the MVP, curated lesion correspondence may be provided through validated source mappings or explicitly reviewed annotations.

---

11. Synthetic Follow-Up Geometry

11.1 Purpose

Use the baseline tumor segmentation and COS-SPEC-004 trajectory values to create visual hypothetical follow-up lesion states.

11.2 Required operation

For each generated time state:

1. Retrieve the authentic baseline lesion mask.
2. Retrieve the target hypothetical tumor measurement.
3. Apply a documented geometric transformation.
4. Measure the resulting synthetic mask.
5. Compare actual resulting geometry with the target.
6. Reject or adjust configurations outside specified tolerance.
7. Generate a synthetic lesion mesh.
8. Store its provenance.

11.3 Simplified isotropic scaling

Where a purely geometric isotropic transformation is used, an approximate linear scale factor for a desired volume ratio is:

Scale = (TargetVolume / BaselineVolume)^(1/3)

This is a geometric relationship, not a biological tumor growth model.

11.4 Advanced shape variations

Optional controlled transformations:

- Mild anisotropic deformation.
- Localized boundary change.
- Irregular growth contours.
- Localized contraction.

These shall remain constrained by geometry-validity rules.

11.5 Synthetic imaging restrictions

Do not fabricate photorealistic follow-up MRI images and label them as genuine clinical scans.

For the initial prototype, synthetic masks and meshes are preferred over generated diagnostic-looking images.

11.6 Synthetic lesion identity

Every synthetic geometry artifact must link to:

- Real source baseline.
- Simulation branch.
- Relative observation date.
- Geometric transformation version.
- Target measurement.
- Actual generated measurement.
- Provenance type.

---

12. Digital Twin State Engine

12.1 State composition

Each time-indexed twin state shall contain:

- Patient identifier.
- Observation or simulation branch identifier.
- Time reference.
- Imaging study reference, where available.
- Lesion segmentation reference.
- 3D mesh reference.
- Tumor measurements.
- Clinical observation references.
- Treatment-event references.
- Provenance classifications.
- Quality and uncertainty metadata.

12.2 State immutability

Adding a new state shall not overwrite a previous published state.

12.3 Branch isolation

Synthetic branch A shall not modify branch B or the authentic patient record.

12.4 State loading

The system shall load a selected state through a defined backend interface.

12.5 State update

A newly imported clinical observation shall trigger only applicable downstream processing.

For example, a new laboratory result should update related clinical summaries without unnecessarily regenerating a tumor mesh.

12.6 Interpolation

Optional interpolation between measured or simulated states may be implemented.

Interpolated states shall use INTERPOLATED provenance and must not be displayed as observed imaging sessions.

---

13. Web-Based Visualization

13.1 Required workspace

The digital twin workspace shall include:

- Interactive 3D tumor viewport.
- Observation-date selector.
- Tumor measurements.
- Clinical timeline.
- Laboratory trend charts.
- Treatment-event markers.
- Provenance and uncertainty indicators.

13.2 3D interaction

Required:

- Rotate.
- Pan.
- Zoom.
- Reset camera.
- Select visit.
- Toggle supported geometry layers.

Optional:

- Opacity control.
- Mesh wireframe.
- Previous/current overlay.
- Side-by-side comparison.
- Tumor cross-sectional slicing.

13.3 Rendering approach

The engineering team may use vtk.js or Three.js for lesion geometry.

OHIF/Cornerstone3D may provide the medical image review surface.

The chosen design should avoid implementing an entire general-purpose radiology viewer from scratch.

13.4 Coordinate consistency

When a 3D tumor mesh is overlaid on MRI data, both must use compatible physical-space transformations.

The application shall not treat voxel index coordinates as millimeter coordinates.

13.5 Visual provenance

Recommended presentation classes:

- Authentic observation.
- Authentic observation with derived measurement.
- Synthetic follow-up geometry.
- Interpolated geometry.
- Predicted geometry.

The UI must include explicit text labels; color alone shall not be relied upon.

13.6 Timeline interaction

Changing observation date shall update:

- Active mesh.
- Tumor measurements.
- Associated MRI study, where available.
- Clinical charts.
- Treatment events.
- Evidence status.

13.7 Unavailable data

If a visit contains only laboratories without new imaging, the UI must not invent an MRI.

It may display the last available imaging state with its actual acquisition date and an explicit indication that no new scan was recorded.

---

14. Backend Component Boundaries

The subsystem shall be separated into services or modules with clearly defined interfaces.

Imaging ingestion

Responsibilities:

- Source validation.
- DICOM/NIfTI loading.
- Metadata extraction.
- Imaging artifact registration.

Imaging preprocessing

Responsibilities:

- Orientation handling.
- Spatial resampling.
- Intensity normalization.
- Model-specific input preparation.

Segmentation

Responsibilities:

- Reference-mask import.
- Model inference.
- Segmentation output validation.
- Version management.

Measurement service

Responsibilities:

- Lesion measurements.
- Physical-space calculation.
- Temporal change calculations.

Registration service

Responsibilities:

- Image alignment.
- Transformation storage.
- Alignment quality checks.

Geometry service

Responsibilities:

- Mask-to-mesh conversion.
- Mesh validation.
- Synthetic geometry generation.

Digital twin state service

Responsibilities:

- Observation-to-geometry linkage.
- Temporal state management.
- Scenario isolation.
- Viewer artifact manifests.

Visualization API

Responsibilities:

- Authorized access to image previews.
- Geometry manifest delivery.
- Tumor measurements.
- Longitudinal state metadata.

---

15. Proposed API Contracts

These are preliminary contracts; COS-SPEC-008 will finalize endpoint names and payload schemas.

POST /api/imaging/studies/import

Import an approved imaging study.

Returns:

- Study ID.
- Import status.
- Validation issues.
- Processing job reference.

POST /api/imaging/studies/{study_id}/preprocess

Initiate approved preprocessing.

POST /api/imaging/studies/{study_id}/segment

Run or import a suitable tumor segmentation.

GET /api/imaging/studies/{study_id}/segmentation

Retrieve segmentation metadata and artifact references.

GET /api/imaging/studies/{study_id}/measurements

Retrieve calculated tumor measurements.

POST /api/imaging/registrations

Initiate registration of compatible studies.

POST /api/geometry/generate

Generate a 3D mesh from a valid mask.

GET /api/twins/{patient_id}/states

Retrieve available time-indexed twin states.

GET /api/twins/{patient_id}/states/{state_id}

Retrieve metadata and authorized artifact references for a specific state.

GET /api/twins/{patient_id}/timeline

Retrieve chronological imaging and clinical state information.

All endpoints must use server-enforced access controls.

Large imaging artifacts should be transferred using controlled storage links or streaming rather than embedding them in clinical JSON responses.

---

16. Evaluation and Quality Assurance

16.1 Preprocessing tests

- Correct source volume dimensions.
- Correct affine and orientation.
- Round-trip coordinate consistency.
- Correct image/mask alignment.
- Valid resampling.
- Preserved source metadata.

16.2 Segmentation tests

- Mask loading correctness.
- Spatial alignment.
- Reference-mask agreement.
- Dice and surface-based metrics where references exist.
- Correct handling of empty or invalid masks.

16.3 Measurement tests

- Known synthetic voxel geometry.
- Known expected physical volume.
- Resolution invariance within tolerance.
- Correct percentage-change calculations.
- Invalid-denominator handling.

16.4 Registration tests

- Identity transformation.
- Known synthetic rigid transformation.

- Successful transform inversion.
- Detection of alignment failure.

16.5 Geometry tests

- Nonempty mesh.
- Valid coordinates.
- Mesh-to-mask spatial alignment.
- Consistent displayed dimensions.
- Appropriate simplification.

16.6 Temporal tests

- Correct study dates.
- Correct lesion associations.
- Missing-scan handling.
- Preserved historical states.
- Branch isolation.

16.7 Visualization tests

- Load and rotate a mesh.
- Change timeline state.
- Synchronize measurements.
- Preserve provenance labels.
- Handle missing geometry.
- Recover gracefully from failed asset loads.

---

17. Proposed Performance Targets

These are internal engineering targets, not measured results.

- Curated mesh rotation: interactive on the target demonstration browser.
- Timeline state switching for preloaded meshes: target under two seconds.

- Patient workspace loading: target under five seconds for preprocessed demonstration assets under suitable network conditions.
- Basic tabular measurement updates: target under one second for local operations.
- Large imaging preprocessing: asynchronous, with progress or job status.
- GPU-heavy segmentation: optional remote worker or precomputed result.

No clinical performance guarantee is implied.

The pipeline shall support bounded-memory processing to accommodate local development hardware with approximately 8 GB RAM.

---

18. Mandatory Acceptance Criteria

AC-IMG-01: Import at least one genuine breast MRI study with correct geometry.

AC-IMG-02: Load a suitable validated tumor segmentation.

AC-IMG-03: Calculate tumor volume from its physical voxel representation.

AC-IMG-04: Render a rotatable and zoomable 3D lesion model.

AC-IMG-05: Associate each geometry artifact with the correct patient and observation date.

AC-IMG-06: Distinguish authentic and synthetic lesion geometry.

AC-IMG-07: Navigate at least four states in a curated synthetic trajectory.

AC-IMG-08: Show at least one genuine patient with multiple actual imaging sessions.

AC-IMG-09: Handle unavailable imaging without creating fabricated scans.

AC-IMG-10: Preserve authentic source files and historical states.

AC-IMG-11: Demonstrate correct physical volume calculations using known test objects.

AC-IMG-12: Detect spatial misalignment or inadequate registration quality.

AC-IMG-13: Maintain lesion and branch identity without cross-patient contamination.

AC-IMG-14: Produce reproducible geometry artifacts from the same validated inputs and transformation settings.

AC-IMG-15: Report actual segmentation evaluation metrics where appropriate reference annotations exist.

---

19. Required Deliverables

1. MRI ingestion module.
2. Imaging metadata validator.
3. DICOM-to-NIfTI processing capability.
4. Standardized preprocessing pipeline.
5. Segmentation-mask import and validation.
6. Automated segmentation adapter, if supported.
7. Physical measurement engine.
8. Registration and quality-check module.
9. Lesion mask-to-mesh generator.
10. Synthetic geometry generator.
11. Temporal lesion-tracking model.
12. Digital twin state service.
13. Authorized visualization artifact API.
14. Interactive 3D viewer.
15. Test fixtures and evaluation reports.
16. Imaging limitations documentation.

---

20. Research and Technical References

- MONAI: https://monai.io/
- MONAI documentation: https://docs.monai.io/
- nnU-Net v2: https://github.com/MIC-DKFZ/nnUNet
- SimpleITK: https://simpleitk.org/
- SimpleITK documentation: https://simpleitk.readthedocs.io/
- DICOM standard: https://www.dicomstandard.org/

- OHIF Viewer: https://docs.ohif.org/
- Cornerstone3D: https://www.cornerstonejs.org/
- vtk.js: https://kitware.github.io/vtk-js/
- TCIA I-SPY2: https://www.cancerimagingarchive.net/collection/ispy2/

---

21. Final Engineering Directive

CancerOS shall treat spatial geometry as scientific data, not merely as visual content.

All imaging-derived measurements must be calculated from documented image and segmentation geometry.

All synthetic anatomical states must be identified as simulations.

The central imaging deliverable is a reliable transformation:

Authentic MRI → validated tumor mask → measured 3D geometry → timestamped digital twin state → longitudinal comparison.

This transformation must work independently of the clinical language model.

End of COS-SPEC-005 — Version 1.0